Supplementary Material for "Special optimal thermal cells: correlator-designed displacements for finite-temperature effective-harmonic phonons"

Albert Linda

Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, India

S1. SURROGATE DESIGN STUDY

Table S1 lists the median phonon-mesh errors of six displacement designs on the DFT-fitted FC2+FC3+FC4 surrogates of Al, W, and NaCl. The surrogate effective FC2 was fitted to 6000 paired canonical samples; fitting half of them changes the mesh frequencies by 0.002 (Al), 0.001 (W), and 0.006 THz (NaCl). All designs were drawn from the harmonic FD model of the same $3 \times 3 \times 3$ cell and fitted in the space-group adapted FC2 basis with the cutoffs of the main text. Held-out forces on 200 canonical configurations follow the same ordering.

S2. DESIGNS IN THE CRYSTAL-SYMMETRY BASIS

Table S2 repeats Table V of the main text with all fits in the FC2 basis of the full crystal space group (hiPhive cluster space, 3–12 parameters, cutoff limited to half the shortest image distance). For Si and Mg the two bases coincide.

Table S3 repeats the DFT sTDEP comparison of the main text in the crystal-symmetry basis.

S3. REGRESSION FRAMEWORKS

Table S4 compares nine regression methods of the trainstation package, as used in hiPhive workflows, on the same DFT data and the same crystal-symmetry cluster space. Hyperparameters were selected by the package's internal cross-validation. All methods agree to within 0.03 THz.

S4. DERIVATIONS FOR THE ESTIMATOR THEORY

Effective FC2 as averaged Hessian. For $\mathbf{u} \sim N(\mathbf{0}, \mathbf{C})$ and smooth V, Stein's lemma states $\langle \mathbf{u}\, g(\mathbf{u}) \rangle = \mathbf{C} \langle \nabla g \rangle$. With $g = \mathbf{F}^{\mathrm{T}} = -\nabla V^{\mathrm{T}}$ this gives $\langle \mathbf{u}\mathbf{F}^{\mathrm{T}} \rangle = -\mathbf{C} \langle \nabla\nabla V \rangle$, so the population least-squares coefficient of $\mathbf{F}$ on $\mathbf{u}$ is $-\langle \mathbf{F}\mathbf{u}^{\mathrm{T}} \rangle \mathbf{C}^{-1} = \langle \nabla\nabla V \rangle$. The SCHA free-energy gradient with respect to the auxiliary FC2 vanishes when $\langle \nabla\nabla V \rangle_{\mathbf{C}(\mathbf{\Phi})} = \mathbf{\Phi}$ for a fixed centroid, which is the self-consistent fixed point of the main text.

Fixed-amplitude bias. In mass-weighted normal coordinates with forces $f_a = -\omega_a^2 Q_a - \frac{1}{2} V_{abc} Q_b Q_c - \frac{1}{6} V_{abcd} Q_b Q_c Q_d$, the regression coefficient for a design with independent symmetric coordinates is $\Phi_{ae} = -\langle f_a Q_e \rangle / \langle Q_e^2 \rangle$. The cubic term vanishes by symmetry. The quartic term needs $\langle Q_b Q_c Q_d Q_e \rangle$, which for distinct indices paired as $(bb)(ee)$ equals $\sigma_b^2 \sigma_e^2$ for both Gaussian and $\pm\sigma$ coordinates, and for $b = c = d = e$ equals $3\sigma_e^4$ (Gaussian) or σ_e^4 (fixed amplitude). Collecting the three pairings,

$$\Phi_{ae} = \omega_a^2 \delta_{ae} + \tfrac{1}{2} \sum_b V_{abbe} \sigma_b^2 - \tfrac{1}{6} \left(3 - \kappa_e\right) V_{aeee} \sigma_e^2, \quad \text{(S1)}$$

with $\kappa_e = \langle Q_e^4 \rangle / \sigma_e^4$. The last term vanishes for Gaussian coordinates ($\kappa_e = 3$) and equals $-\frac{1}{3} V_{aeee} \sigma_e^2$ for the fixed-amplitude design ($\kappa_e = 1$). The statement is verified numerically by exact enumeration of all 2^6 sign patterns of a random six-mode quartic model in the test suite (`test_optimal_design.py`), as are the exact cancellation of arbitrary cubic forces by $\pm$ pairs and the thermal mean-square displacements of the ZG and rattle designs.

S5. HIGHER-ORDER FORCE-CONSTANT VALIDATION

Figure S1 documents the joint FC2+FC3+FC4 fits used as surrogates: held-out errors of the nested models, convergence with the number of training snapshots, sensitivity to the interaction cutoffs, and comparison with the force changes produced by higher plane-wave cutoffs and denser k meshes. For Al the denser-k-mesh force change exceeds the final fit error, so the Al tensors are used here only as model potentials for the sampling study, not as converged inputs for transport calculations.

S6. INDIVIDUAL ITERATIONS OF THE SOTC-C RUNS

TABLE S1. Surrogate design study (Al 300 K, W 1500 K, NaCl 700 K; DFT-fitted FC2+FC3+FC4 surrogates in $3 \times 3 \times 3$ cells). Median mesh RMSE (THz) against the surrogate effective FC2 over 30 repetitions, at 2/4/8/16 force calls.

Design	Al	W	NaCl
Canonical	0.199/0.123/0.088/0.064	0.091/0.068/0.048/0.033	0.224/0.185/0.116/0.089
Canonical ±	0.066/0.054/0.038/0.032	0.035/0.027/0.021/0.018	0.176/0.127/0.078/0.078
Exact ampl., random signs	0.066/0.050/0.038/0.025	0.035/0.028/0.017/0.016	0.166/0.131/0.093/0.063
SOTC-D	0.063/0.056/0.040/0.025	0.039/0.024/0.021/0.015	0.150/0.131/0.090/0.062
Stratified ampl. + D-opt.	0.073/0.046/0.038/0.028	0.035/0.031/0.020/0.015	0.151/0.134/0.088/0.089
Gaussian ampl. + D-opt.	0.074/0.053/0.046/0.030	0.048/0.032/0.024/0.019	0.184/0.124/0.114/0.077
Harmonic vs effective	0.229	0.149	0.765

TABLE S2. Updated trial (iteration 1, crystal basis). Mesh RMSE (THz) against the iteration-1 effective FC2. Matched trial: SOTC-D designed from the iteration-1 trial (D_1) versus the better of random canonical subsets and pairs drawn from the same trial (C_1), scored against the disjoint effective FC2. Equal cost: bootstrapped SOTC-D (B; first pair from the harmonic trial, later pairs from the model refitted to the first pair, no extra force calls) and one-shot SOTC-D (D_0) versus canonical pairs from the harmonic trial (C_0; a single sTDEP iteration), scored against the 40-snapshot iteration-1 effective FC2, which shares no configuration with them. B and D_0 coincide at 2 calls by construction. The best entry of each group is bold.

	matched, 2		equal cost, 2			matched, 8		equal cost, 8		
	D_1	C_1	B	D_0	C_0	D_1	C_1	B	D_0	C_0
Al	**0.06**	0.08	**0.07**	0.07	0.10	**0.03**	0.05	**0.03**	0.05	0.04
Ag	**0.04**	0.11	**0.02**	0.02	0.13	0.06	**0.05**	**0.06**	0.09	0.10
Au	0.08	**0.08**	**0.15**	0.15	0.24	**0.05**	0.05	**0.02**	0.17	0.22
Cu	**0.08**	0.15	0.20	0.20	**0.07**	**0.04**	0.08	0.17	**0.10**	0.14
Mo	0.06	**0.06**	**0.05**	0.05	0.07	**0.02**	0.03	**0.02**	0.03	0.06
W	0.05	**0.04**	0.05	0.05	**0.04**	**0.02**	0.02	0.02	**0.02**	0.02
NaCl	**0.05**	0.11	**0.11**	0.11	0.14	0.09	**0.07**	**0.05**	0.15	0.12
MgO	**0.22**	0.32	0.42	0.42	**0.34**	**0.09**	0.15	**0.11**	0.32	0.26
PbTe	**0.02**	0.07	0.19	0.19	**0.13**	**0.02**	0.03	**0.03**	0.12	0.09
Si	**0.07**	0.10	**0.10**	0.10	0.11	**0.05**	0.05	**0.05**	0.09	0.07
Mg	**0.05**	0.06	**0.06**	0.06	0.07	**0.03**	0.05	**0.02**	0.06	0.04
$D_1 < C_1$ / $B < C_0$	8/11		7/11			9/11		9/11		

TABLE S5: Individual iterations of the SOTC-C runs: configurations, r.m.s. displacement, unstable mesh fraction and minimum frequency of the axial-prior fit, and normalized condition number of the space-group adapted design.

Material	iter.	N_{ens}	u_{rms} (Å)	f_{unst}	ω_{min} (THz)	cond.
Al	1	5	0.106	0.000	0.60	5.2
Al	2*	5	0.106	0.000	0.48	3.4
Ag	1	5	0.167	0.000	0.26	5.7
Ag	2	5	0.167	0.000	0.42	3.2
Ag	3	5	0.167	0.000	0.44	3.2
Ag	4*	5	0.167	0.000	0.32	2.8
Au	1*	1	0.186	0.007	-0.95	10.7
Au	2	1	0.186	0.159	-4.58	7.5
Cu	1	8	0.169	0.000	0.51	15.3
Cu	2	8	0.169	0.000	0.45	7.9
Cu	3*	8	0.169	0.000	0.45	7.5
Cu	4	8	0.169	0.000	0.52	8.2
Mo	1	1	0.111	0.000	0.52	17.1
Mo	2*	1	0.111	0.000	0.99	8.0
W	1	3	0.115	0.000	0.76	4.5
W	2*	3	0.115	0.000	0.73	2.9
W	3	3	0.115	0.000	0.75	3.5
W	4	3	0.115	0.000	0.74	3.3
NaCl	1	3	0.191	0.000	0.31	3.9
NaCl	2	3	0.191	0.000	0.34	2.8
NaCl	3	3	0.191	0.000	0.32	2.9
NaCl	4	3	0.191	0.000	0.39	3.1
NaCl	5*	3	0.191	0.000	0.31	2.4
MgO	1	3	0.141	0.000	0.96	2.9
MgO	2	3	0.141	0.000	0.73	2.1

TABLE S3. Self-consistent protocols with DFT forces, equal cost. Mesh RMSE (THz) against the 40-snapshot iteration-1 effective FC2 (crystal basis). B: bootstrapped SOTC-D. sTDEP: half of the calls as canonical ± pairs from the harmonic trial, half as canonical ± pairs from the model refitted to them, fitted to the second iteration only (last) or to both (pool); median of three seeds. C_0: one-shot canonical pairs from the harmonic trial (median of 20 subsets). No configuration of any protocol enters the reference. The best entry of each group is bold.

	8 calls				16 calls			
	B	sTDEP last	sTDEP pool	C_0	B	sTDEP last	sTDEP pool	C_0
Al	**0.03**	0.05	0.05	0.04	0.02	0.03	**0.01**	0.03
Ag	0.06	**0.04**	0.07	0.10	0.04	**0.03**	0.09	0.10
Au	**0.02**	0.06	0.17	0.22	**0.01**	0.04	0.15	0.21
Cu	0.17	0.18	**0.10**	0.14	0.16	0.08	**0.06**	0.14
Mo	**0.02**	0.05	0.03	0.06	0.02	0.03	**0.02**	0.05
W	0.02	0.05	0.03	**0.02**	0.02	0.03	**0.02**	0.02
NaCl	**0.05**	0.13	0.09	0.12	**0.04**	0.10	0.13	0.12
MgO	**0.11**	0.22	0.18	0.26	**0.10**	0.19	0.22	0.25
PbTe	**0.03**	0.05	0.07	0.09	**0.04**	0.04	0.06	0.08
Si	**0.05**	0.08	0.08	0.07	**0.04**	0.05	0.06	0.06
Mg	**0.02**	0.05	0.03	0.04	**0.02**	0.04	0.04	0.03
B best		8/11				6/11		

TABLE S4. Regression frameworks (hiPhive cluster space with crystal symmetry, trainstation fit methods) on the same DFT data, 8 force calls. Entries: mesh RMSE (THz) against the disjoint effective FC2 (supercell basis, so the entries include the basis difference listed in the main text) for random canonical subsets (median of 10) / SOTC-D.

Method	Al	Ag	Au	Cu	Mo	W	NaCl	MgO	PbTe	Si	Mg
least-squares	0.20/0.18	0.11/0.09	0.12/0.13	0.23/0.20	0.09/0.08	0.06/0.05	0.13/0.11	0.31/0.26	0.06/0.11	0.10/0.05	0.08/0.04
ridge	0.20/0.18	0.11/0.09	0.12/0.13	0.24/0.20	0.09/0.08	0.06/0.05	0.13/0.12	0.31/0.26	0.06/0.12	0.10/0.05	0.08/0.04
bayesian-ridge	0.20/0.18	0.11/0.09	0.12/0.14	0.24/0.20	0.09/0.08	0.06/0.05	0.13/0.12	0.31/0.26	0.06/0.12	0.10/0.05	0.08/0.03
ardr	0.20/0.18	0.11/0.09	0.12/0.14	0.24/0.20	0.09/0.08	0.06/0.05	0.14/0.11	0.32/0.27	0.07/0.12	0.08/0.06	0.07/0.04
rfe	0.20/0.18	0.11/0.09	0.12/0.13	0.23/0.20	0.09/0.08	0.06/0.05	0.13/0.11	0.33/0.26	0.06/0.12	0.10/0.07	0.07/0.04
omp	0.20/0.18	0.11/0.09	0.12/0.13	0.23/0.20	0.09/0.08	0.06/0.05	0.14/0.11	0.31/0.26	0.06/0.11	0.10/0.08	0.07/0.04
lasso	0.20/0.18	0.11/0.09	0.12/0.13	0.23/0.20	0.09/0.08	0.06/0.05	0.14/0.11	0.31/0.26	0.06/0.11	0.09/0.06	0.07/0.04
adaptive-lasso	0.20/0.18	0.13/0.12	0.12/0.16	0.23/0.20	0.09/0.08	0.06/0.05	0.14/0.11	0.31/0.26	0.06/0.11	0.09/0.05	0.08/0.06
elasticnet	0.20/0.18	0.11/0.09	0.12/0.13	0.23/0.20	0.09/0.08	0.06/0.05	0.14/0.11	0.31/0.26	0.06/0.11	0.09/0.06	0.07/0.04

Material	iter.	N_{ens}	u_{rms} (Å)	f_{unst}	ω_{min} (THz)	cond.
MgO	3	3	0.141	0.000	0.96	2.6
MgO	4*	3	0.141	0.000	0.88	2.2
PbTe	1	5	0.190	0.000	0.21	3.1
PbTe	2*	5	0.190	0.000	0.22	2.2
PbTe	3	5	0.190	0.000	0.19	2.1
PbTe	4	5	0.190	0.000	0.22	2.4

An asterisk marks the iteration selected by the workflow. The two Au iterations each contain a single configuration; their axial-prior fits are unstable, whereas the pooled fit is stable.

S7. LATTICE-PARAMETER SENSITIVITY

S8. ACCURACY OVERVIEW

S9. NACL OPTIC MODES AND NEUTRON DATA

S10. THERMAL DISPLACEMENT CORRELATOR

In second quantization, the displacement of basis atom a in cell $\mathbf{R}$ is

$$\hat{u}_a^\alpha(\mathbf{R}) = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{q}s} \sqrt{\frac{\hbar}{2M_a\omega_{\mathbf{q}s}}}\, e_a^\alpha(\mathbf{q}s) \left(\hat{a}_{\mathbf{q}s} + \hat{a}^\dagger_{-\mathbf{q}s}\right) e^{i\mathbf{q}\cdot\mathbf{R}}. \tag{S2}$$

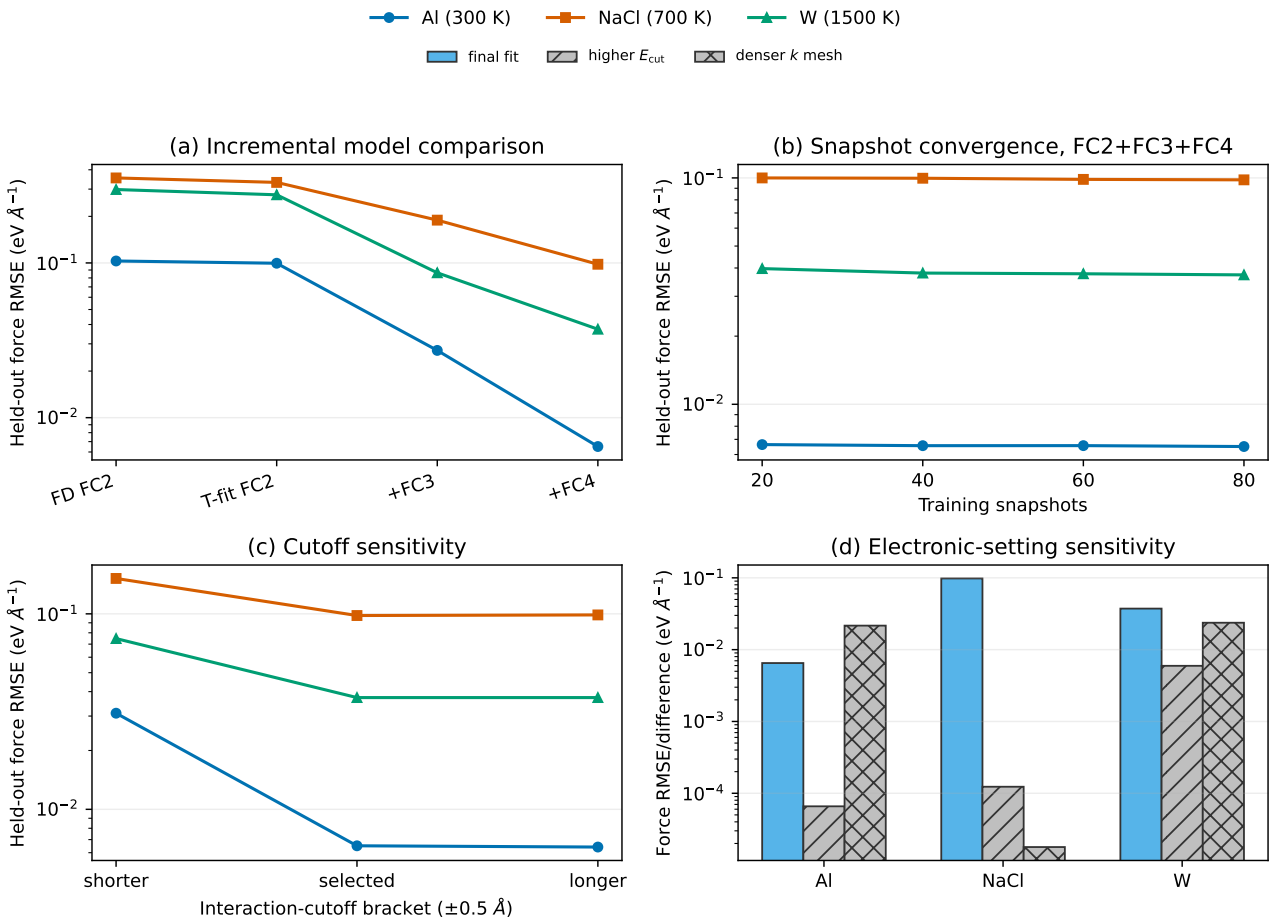


FIG. S1. Validation of the FC2+FC3+FC4 fits to 80 canonical QE snapshots (Al 300 K, W 1500 K, NaCl 700 K; $3 \times 3 \times 3$ cells). (a) Held-out force RMSE of harmonic FD FC2 and of temperature-fitted FC2, FC2+FC3, and FC2+FC3+FC4 models. (b) Convergence with training snapshots. (c) Cutoff sensitivity (±0.5 Å). (d) Fit error compared with electronic-setting force differences.

TABLE S6. PBE equilibrium lattice parameters from Birch–Murnaghan fits (primitive cells) and the effect of using them instead of the production values on the harmonic $3 \times 3 \times 3$ FD model: mesh RMSE and change in $S_{\mathrm{vib}}(T_{\mathrm{des}})$.

	a_{prod} (Å)	a_{PBE} (Å)	$\Delta a/a$ (%)	B_0 (GPa)	RMSE (THz)	ΔS_{vib}
Al	4.050	4.039	-0.26	78	0.10	-0.38
Ag	4.120	4.146	+0.63	91	0.16	+1.17
Au	4.130	4.150	+0.49	142	0.12	+1.17
Cu	3.670	3.631	-1.06	142	0.35	-1.71
Mo	3.160	3.160	-0.01	259	0.00	-0.01
W	3.180	3.184	+0.14	305	0.03	+0.18
NaCl	5.730	5.704	-0.45	24	0.12	-0.64
MgO	4.260	4.249	-0.25	153	0.14	-0.30
PbTe	6.510	6.567	+0.88	39	0.14	+1.72

With $\langle \hat{a}^\dagger_{\mathbf{q}s} \hat{a}_{\mathbf{q}'s'} \rangle = n_B \delta\delta$, $\langle \hat{a}_{\mathbf{q}s} \hat{a}^\dagger_{\mathbf{q}'s'} \rangle = (n_B+1)\delta\delta$, vanishing anomalous averages, and $2n_B + 1 = \coth(\hbar\omega/2k_BT)$, the canonical average $\langle \hat{u}^\alpha_a(\mathbf{0}) \hat{u}^\beta_b(\mathbf{R}) \rangle$ reduces to Eq. (4) of the main text: only $\mathbf{q}' = -\mathbf{q}$ contributes, and $e^\beta_b(-\mathbf{q}s) = e^{\beta *}_b(\mathbf{q}s)$ gives the factor $e^\alpha_a(\mathbf{q}s) e^{\beta *}_b(\mathbf{q}s) e^{-i\mathbf{q}\cdot\mathbf{R}}$. Its trace with the completeness relation of the polarization vectors gives $U_{\mathrm{iso}} = (\hbar/6N_cM) \sum_{\mathbf{q}s} \omega^{-1}_{\mathbf{q}s} \coth(\hbar\omega_{\mathbf{q}s}/2k_BT)$ for a monatomic crystal.

In the commensurate supercell the same covariance is

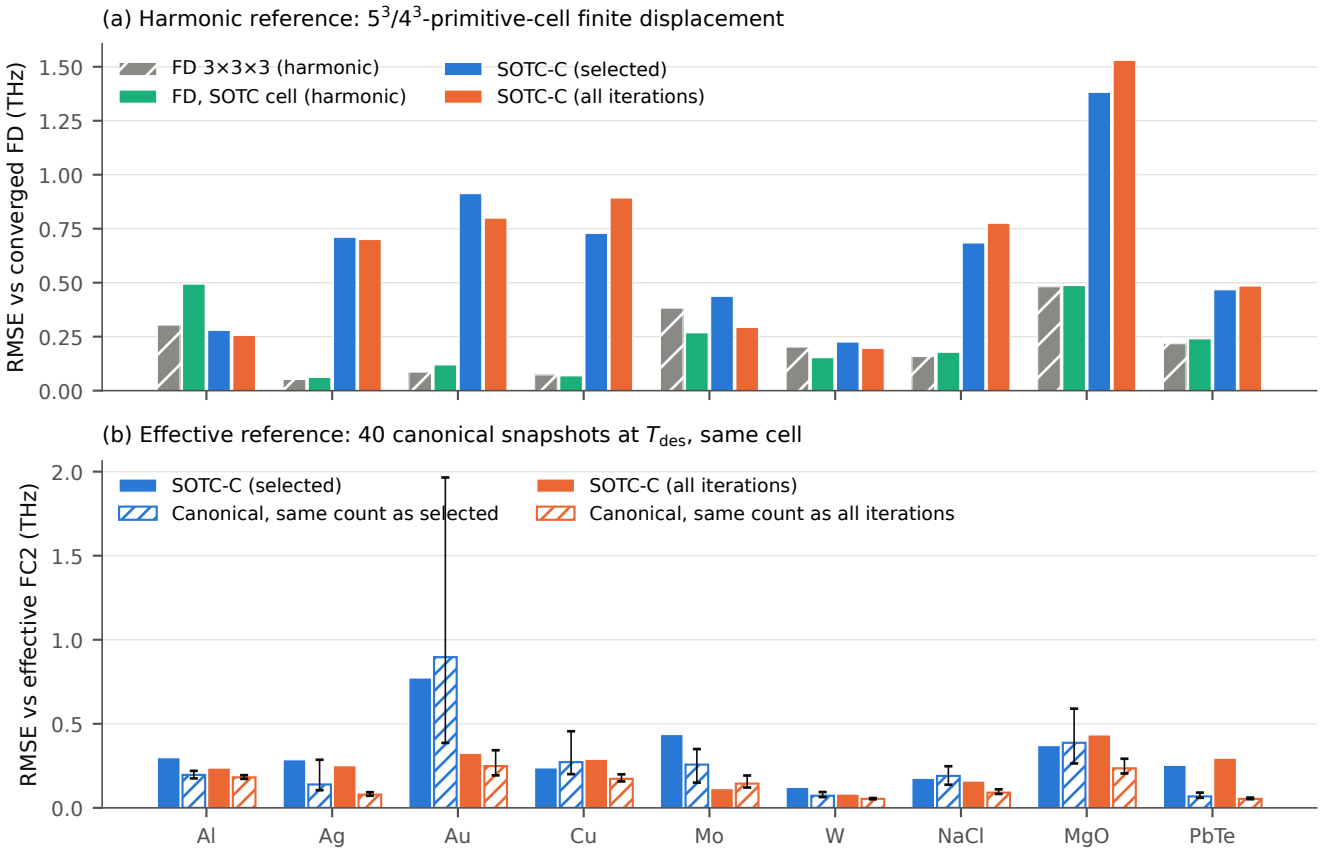


FIG. S2. (a) Mesh RMSE against the converged large-cell harmonic FD reference for harmonic FD in $3 \times 3 \times 3$ cells and in the SOTC cells, and for the selected and pooled SOTC-C models. (b) Mesh RMSE against the same-cell effective FC2 for the selected and pooled SOTC-C models and for random canonical subsets of the same size fitted with the same prior (median and 10–90% range over 20 draws).

diagonal in the normal-mode coordinates Q_m with variances σ_m^2 given in Sec. II D of the main text. A configuration of the form of Eq. (7) has $Q_m^2 = \sigma_m^2$ for every mode, so that its single-configuration second moment equals the canonical one mode by mode; cross terms $Q_m Q_{m'} = s_m s_{m'} \sigma_m \sigma_{m'}$ average to zero only over the sign patterns, and their effect on the force fit is what the D-optimal sign choice controls.

S11. SOTC OBJECTIVE GRADIENT

For the shell objective of Eq. (6) of the main text, write the cell correlator as a sum over ordered atom pairs, $\bar{C}^{\mathrm{cell}}_{2,ab}(R;\mathbf{u}) = \sum_{i,j} P_{ij}^{abR}\, \mathbf{u}_i \cdot \mathbf{u}_j$, where the symmetric weight matrix P_{ij}^{abR} equals $1/(3N_R^{ab}N_a)$ (or half of it for each order when $a \neq b$) if the pair belongs to shell R of the species channel and zero otherwise; the on-site term is the shell $R = 0$. Then

$$\frac{\partial \mathcal{Q}}{\partial \mathbf{u}_i} = \sum_{a,b,R} 4\, w_{ab}(R)\, \Delta_{ab}(R) \sum_j P_{ij}^{abR}\, \mathbf{u}_j, \qquad \text{(S3)}$$

with $\Delta_{ab}(R) = \bar{C}^{\mathrm{cell}}_{2,ab} - \bar{C}_{2,ab}$, and the mode-amplitude gradient follows by the chain rule through $\mathbf{u}_i = \sum_m A_m \mathbf{e}_{im}/\sqrt{M_i}$. The optimization is carried out in mode amplitudes and phases with random restarts.

[1] G. Raunio, L. Almqvist, and R. Stedman, "Phonon dispersion relations in NaCl," Phys. Rev. **178**, 1496–1501 (1969).
[2] National Bureau of Standards, "Molecular dynamics and structure of solids," NBS Special Publication 301 (National Bureau of Standards, Washington, DC, 1969).

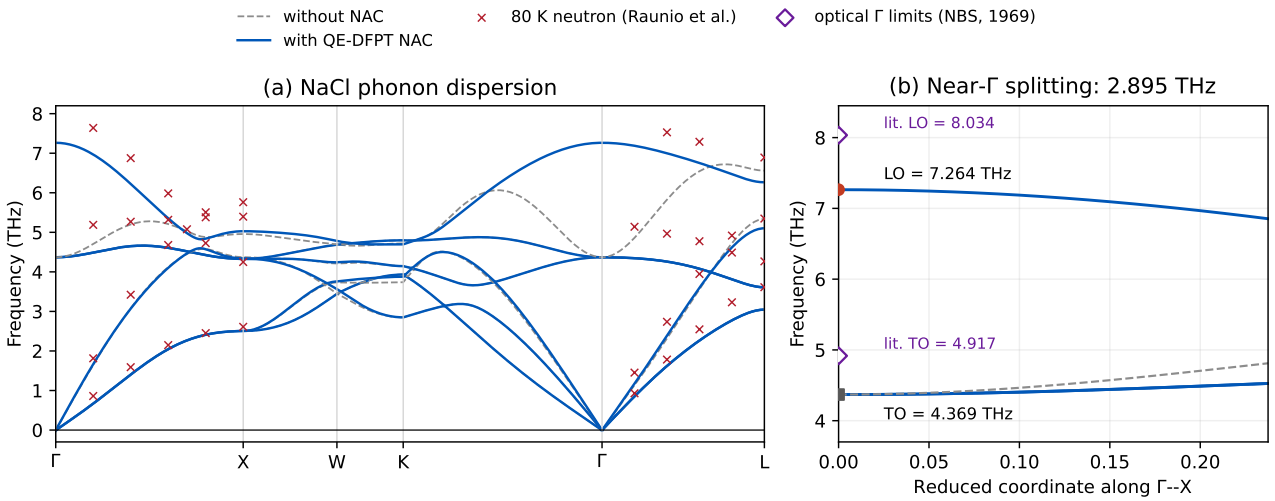


FIG. S3. NaCl harmonic FD FC2 ($3 \times 3 \times 3$ cell) without (gray dashed) and with (blue) the non-analytic correction from QE-DFPT Born charges and dielectric tensor. Red crosses: 39 neutron points at 80 K along Γ–X and Γ–L[1]; purple diamonds: infrared TO and LO limits[2]. The neutron data are temperature-mismatched to the fixed-volume calculation and are shown without a global error metric.